

Data Transformation Techniques For Handling Skewness In Machine Learning

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Introduction

Skewness is a measure of the asymmetry of a distribution. A distribution is asymmetrical when its left and right side are not mirror images.

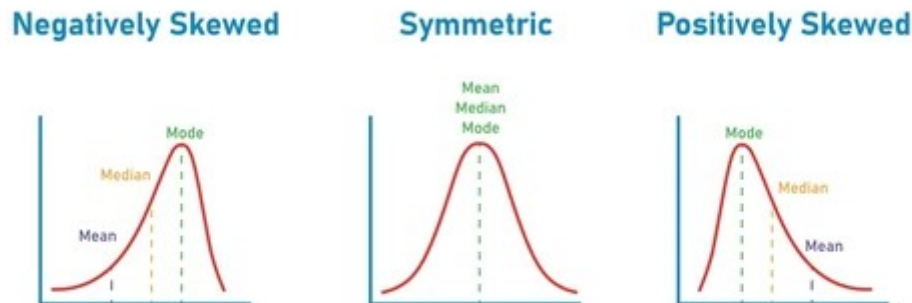


Figure 1: Skewness Types

As shown in figure 1 a distribution can have right (or positive), left (or negative), or zero skewness. A right-skewed distribution is longer on the right side of its peak, and a left-skewed distribution is longer on the left side of its peak:

Understanding the skewness of data is crucial for different reasons:

1. **Modeling Assumptions:** Some machine learning algorithms assume that numerical features have a roughly symmetric distribution. When features are highly skewed, extreme values can have a large impact on the model, reducing its accuracy.
2. **Feature Engineering:** Skewed data can significantly affect the performance of machine learning models. Measuring skewness helps identify features that may benefit from transformations, making the data more suitable for modeling.
3. **Prediction Accuracy:** Highly skewed features can make it more difficult for a model to learn meaningful patterns, reducing its ability to generalize to unseen data. Reducing skewness often improves prediction accuracy by creating a more balanced feature distribution.
4. **Outlier Detection:** Skewed distributions often indicate the presence of extreme values or outliers. Understanding the direction and degree of skewness helps identify where outliers are likely to occur and whether additional preprocessing is needed.

Understanding Skewness

I. Positive Skewness (Right):

A positively skewed distribution is one where most of the data is grouped on the left side, with a long tail stretching out to the right. In this type of distribution, the mean, median, and mode are all positive values. It's also called a right-skewed distribution because the mean tends to be on the right side of the median.

As shown in Figure 2 the mean is higher than the median. This occurs because the data leans toward lower values, which pulls the mean in that direction, while the median stays closer to the center of the data.

Example of positive skewed data: Distribution of scores in a hard exam, The distribution of scores on any particularly difficult exam will be positively skewed with most students scoring around some mean value with a few outlier students scoring much higher.

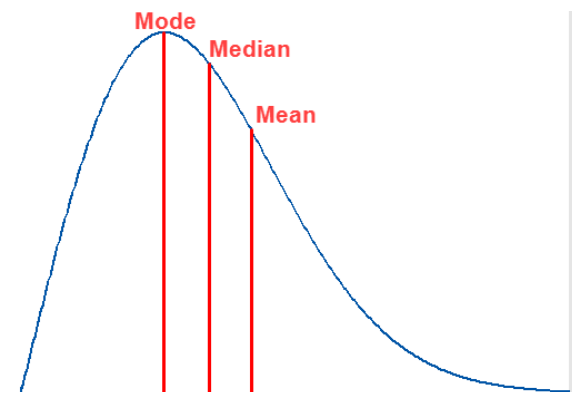


Figure 2: Positively Skewed Data

II. Negative Skewness (Left):

A negatively skewed distribution, also known as a left-skewed distribution, is one where most of the data is clustered on the right side of the graph, with a long tail extending to the left. In this distribution, the mean, median, and mode are typically lower, with the mean being the lowest value. This distribution indicates that the majority of the data points are higher, with fewer lower values pulling the tail to the left.

Example of negative skewed data: Distribution of Olympics Long Jumps In most years, the distribution of long jump lengths for competitors in the Olympics is negatively skewed because most competitors land a jump around 7.5-8 meters, with a few landing a jump of just 5-6 meters.

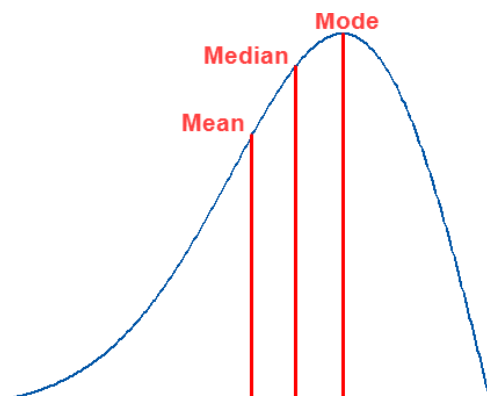


Figure 3: Negatively Skewed Data

Why Skewness Can Be Harmful

Skewed data can negatively affect statistical analysis and the performance of machine learning models. Since skewed distributions contain extreme values and are not symmetric, they may violate the assumptions of some algorithms and reduce prediction accuracy. Therefore, identifying and reducing skewness is an important step in data preprocessing.

- **Effect On Statistical Analysis**

Skewed distributions can distort common statistical measures such as the mean and standard deviation. In a positively skewed distribution, the mean is pulled toward the larger values, while in a negatively skewed distribution, it shifts toward the smaller values. Consequently, these measures may not accurately represent the center and spread of the data. In addition, many statistical methods assume that the data follows an approximately normal distribution, and high skewness may reduce the reliability of their results.

- **Effect On Distance-Based Algorithms**

Distance-based machine learning algorithms, such as K-Nearest Neighbors (KNN) and K-Means clustering, calculate the distance between data points to make predictions or form clusters. When a

feature is highly skewed, extreme values can dominate these distance calculations, causing observations to appear farther apart than they actually are. This may reduce the accuracy of clustering and classification results.

- **Effect On Linear Models**

Linear models, such as Linear Regression and Logistic Regression, are sensitive to extreme values. Highly skewed features may allow a small number of observations to have a disproportionate influence on the fitted model. This can make it more difficult for the model to learn the true relationship between variables, resulting in less accurate predictions. Applying suitable transformations can help reduce skewness and improve model stability.

- **Outlier Detection**

Skewed distributions often indicate the presence of outliers or extreme observations. These values can significantly influence statistical analysis and model training, leading to biased parameter estimates and reduced predictive performance. Understanding the direction and degree of skewness helps identify potential outliers and determine whether additional preprocessing or transformation is required.

- **Effect On Model Performance**

Highly skewed data can reduce the overall performance of machine learning models by making it more difficult to identify meaningful patterns. Features with large skewness may contribute unequally during training, leading to lower prediction accuracy and weaker generalization on unseen data. Reducing skewness through appropriate transformation techniques creates more balanced feature distributions, often resulting in improved model performance.

Detecting Skewness

1. Data Visualization

Visual inspection is one of the simplest ways to detect skewness in numerical features. Plots such as histograms, Kernel Density Estimation (KDE) plots, and boxplots reveal the shape of the data distribution and help identify whether it is symmetric, positively skewed, negatively skewed, or contains extreme values (outliers).

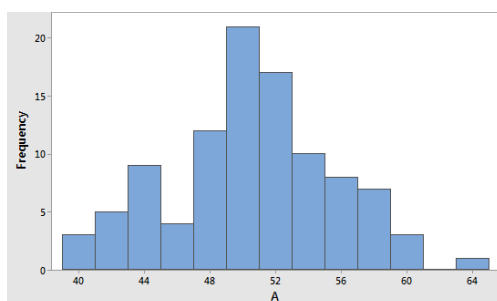


Figure 4: Histogram

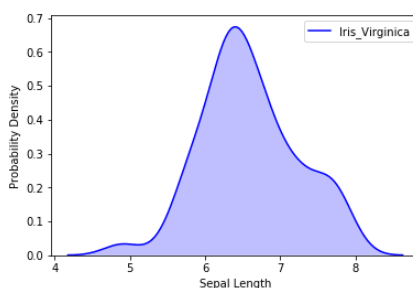


Figure 5: KDE Plot

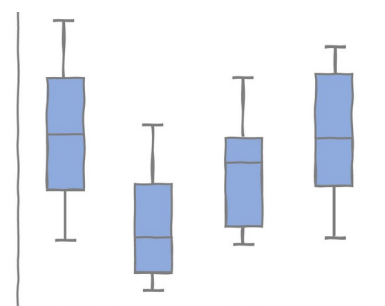


Figure 6: Box Plot

2. Calculating The Coefficient Of Asymmetry

In addition to visual methods, skewness can be measured numerically using the **skewness coefficient**, which quantifies the direction and degree of asymmetry in a distribution. Unlike visual inspection, the skewness coefficient provides an objective measure that allows different features to be compared easily.

$$\text{Formula : Skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

where:

- n = number of observations
- x_i = the i -th observation
- $\bar{x}$ = sample mean
- s = sample standard deviation

Skewness Coefficient	Interpretation
=0	Perfectly Symmetric
Between -0.5 and 0.5	Approximately Symmetric
Between -1 and -0.5 or 0.5 and 1	Moderately Skewed
Greater than 1 or less than -1	Highly Skewed

3. Calculating Mean, Median, and Percentiles

Another useful method for detecting skewness is to compare the **mean**, **median**, and **percentiles** of a numerical feature. In a perfectly symmetric (normal) distribution, the mean and median are approximately equal, and the percentiles are evenly distributed around the median. However, in skewed distributions, extreme values pull the mean toward the longer tail, creating a noticeable difference between the mean and median.

Percentiles provide additional insight into the spread of the data by indicating the values below which a certain percentage of observations fall. Comparing the 25th percentile (Q1), the 50th percentile (median), and the 75th percentile (Q3) helps identify asymmetry in the distribution. Large differences between these values, especially when combined with a gap between the mean and median, may indicate the presence of skewness and outliers.

Handling Positive Skewness

Positive skewness occurs when a distribution has a long tail extending toward larger values. Several transformation techniques can be used to reduce this asymmetry and produce a distribution that is closer to normal.

1. Log Transformation

The logarithmic transformation compresses large values more than small values, making it one of the most effective methods for reducing strong positive skewness.

Formula: $x = \log(x+1)$

- **Advantages**

- ✓ Very effective for highly positively skewed data.
- ✓ Reduces the influence of extreme values.
- ✓ Often improves model performance.

- **Disadvantages**

- ✗ Cannot be applied directly to negative values.
- ✗ Less suitable when the data contains many zeros without using $x+1$ instead of x

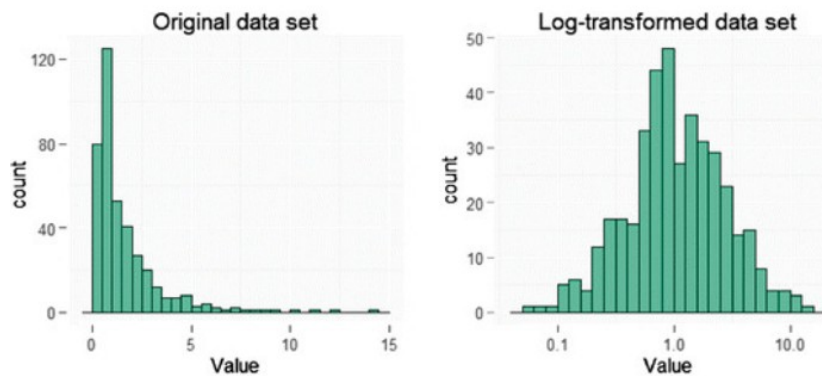


Figure 7: Log Transformation Before and After Histogram

2. Square Root Transformation

The square root transformation reduces positive skewness by compressing larger values while preserving the order of the data. It is less aggressive than the logarithmic transformation.

Formula: $x = \sqrt{x}$

- **Advantages**

- ✓ Simple to apply.
- ✓ Suitable for moderately skewed data.
- ✓ Works well for count data.

- **Disadvantages**

- ✗ Less effective for highly skewed distributions.
- ✗ Requires non-negative values.

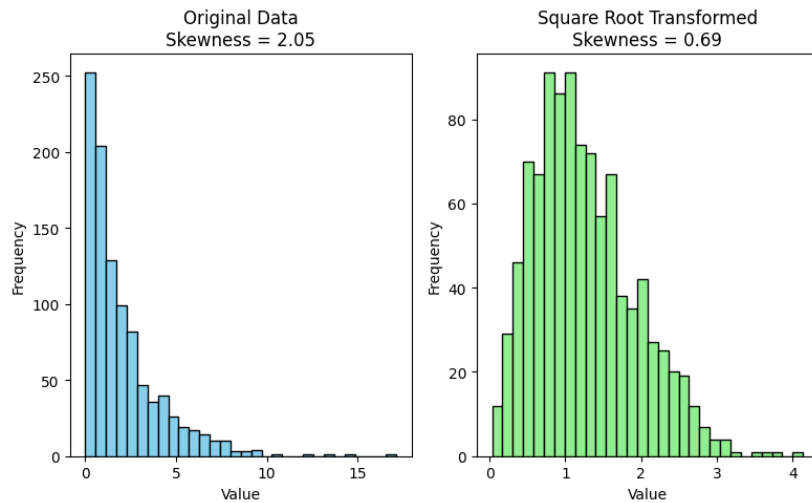


Figure 8: Square Root Transformation Before and After Histogram

3. Cube Root Transformation

The cube root transformation provides a milder reduction in skewness than the logarithmic transformation and can be applied to both positive and negative values.

Formula: $x = x^{\frac{1}{3}}$

- **Advantages**
 1. Can handle zero and negative values.
 - ✓ Produces a moderate reduction in skewness.
- **Disadvantages**
 - ✗ May not sufficiently reduce severe skewness.

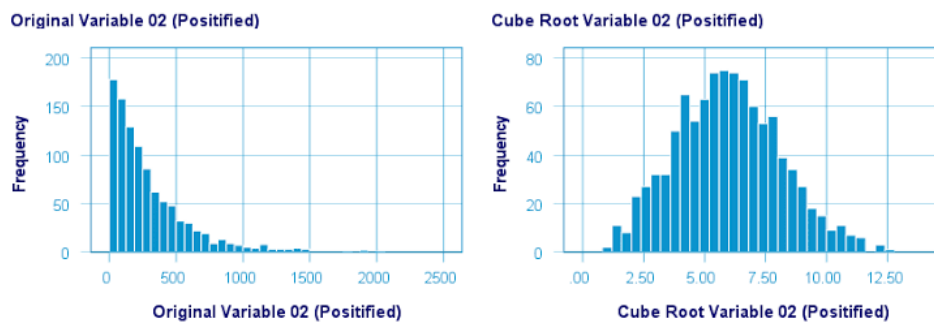


Figure 9: Cube Root Transformation Before and After Histogram

4. Box-Cox Transformation

The Box-Cox transformation automatically determines the most appropriate power transformation to make the data as close to a normal distribution as possible.

- **Advantages**
 - ✓ Often produces the best normalization results.
 - ✓ Improves the performance of many statistical and machine learning models.

- **Disadvantages**

- ✗ Requires all values to be strictly positive.
- ✗ Slightly more computationally expensive than simpler transformations

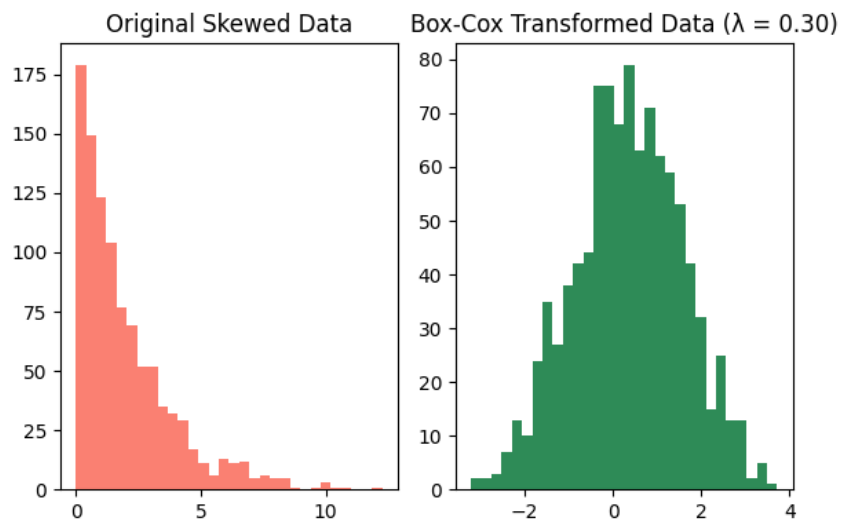


Figure 10: Box-Cox Transformation Before and After Histogram

Handling Negative Skewness

Negative skewness occurs when a distribution has a long tail toward smaller values. A common approach is to **reflect** the data, converting the negatively skewed distribution into a positively skewed one. Once reflected, the same transformation techniques discussed for positive skewness, such as the logarithmic, square root, cube root, and Box-Cox transformations, can be applied to reduce skewness.

The reflection transformation is given by: $x = \max(x) + 1 - x$

Another widely used alternative is the **Yeo-Johnson transformation**, which can handle positive, zero, and negative values directly without requiring reflection.

- **Yeo-Johnson transformation Advantages:**

- ✓ Handles all numerical values directly.
- ✓ Often produces distributions close to normal.

- **Yeo-Johnson transformation Limitations:**

- ✗ More computationally complex than basic transformations.

Box-Cox vs. Yeo-Johnson

Feature	Box-Cox	Yeo-Johnson
Data Requirement	Positive values only	Positive, zero, and negative values
Advantages	Effectively reduces skewness and often achieves near-normal distributions	Works with all numerical values without reflection
Disadvantages	Cannot be applied to zero or negative values	Slightly more computationally complex
Suitable Use Case	Positive numerical features	Features containing positive, zero or negative values

The Box-Cox transformation is preferred when all feature values are positive, while the Yeo-Johnson transformation is more flexible because it can be applied to any numerical feature without requiring data reflection.